Sheet (2)

Answer the following question:

a) Mention the different services offered by operating system to be helpful to the user.

1. User Interface (CLI, GUI, Batch)
2. Program Execution (Loading and running programs)
3. I/O Operations (Handling input/output devices)
4. File-System Manipulation (Creating, deleting, reading, writing files)
5. Communications (Inter-process and network communication)
6. Error Detection (Handling hardware/software errors)

b) Explain the role and importance of system calls in an operating system. Describe the process of how a system call is executed, including the transition between user mode and kernel mode.

A system call is a mechanism that allows a user-space program to request services from the operating system's kernel, such as file operations, process control, or device management. Since user programs cannot directly access hardware or critical system resources, system calls provide a safe and controlled interface between user mode and kernel mode.

-Role and Importance:

- Bridge between user and kernel: System calls enable communication between user applications and the OS kernel.
- Security and protection: They ensure that user programs cannot perform unauthorized or unsafe operations on system resources.
- Controlled resource access: The OS manages hardware and memory through system calls, maintaining system stability.
- Ease of programming: Applications can use high-level APIs (like C library or Win32 API), which internally use system calls to perform operations such as open(), read(), or write().

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Execution Process of a System Call (User Mode to Kernel Mode Transition)

User Request:
A user program calls a library function such as open() to access a file.

System Call Interface:
The function translates the request into a system call number that identifies the required kernel service.

Mode Switch – User Mode to Kernel Mode:
A software interrupt (trap) instruction is executed (for example, int 0x80), causing the CPU to switch from user mode (restricted access) to kernel mode (full access).

Kernel Execution:
The OS uses the system call number to locate the appropriate function in the system call table, then executes the corresponding kernel routine.
Example for open():

1. Checks file permissions and existence.

2. Allocates a file descriptor.

3. Loads file metadata into memory.

Return to User Mode:
After the kernel finishes the operation, it returns the result (success code, error code, or data) to the user program, and the CPU switches back to user mode.

c) Explain the difference between user goals and system goals in the context of operating systems design.

User Goals: Convenience, ease of use, reliability, speed, and safety.
System Goals: Efficiency, ease of design, implementation, maintenance, flexibility, and error free operation.

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d) What is the difference between policy and mechanism? Is separation between them is one of the most important operating systems design principles? If so, what the advantage of the separation?

Policy: Determines what will be done (e.g., scheduling algorithm).
Mechanism: Determines how to do it (e.g., timer interrupts for scheduling).
Advantage of Separation: Flexibility - policies can be changed without modifying mechanisms.

e) Compare and contrast the monolithic and microkernel operating system structures. Discuss the advantages and disadvantages of each

Feature	Monolithic Kernel	Microkernel
Structure	All OS services (file, process, I/O, etc.) run together in kernel mode	Only essential functions (scheduling, IPC) in kernel; other services run in user mode
Examples	MS-DOS, UNIX, early Linux	macOS (Darwin), QNX, Minix
Performance	Very fast (direct function calls)	Slower (message passing between processes)
Security & Stability	Less secure – any bug can crash the system	More secure – faults isolated to user-mode services
Maintenance	Hard to modify or debug (tightly coupled)	Easier to extend or update (modular design)
Design Complexity	Simple to design but hard to maintain	More complex design but easier maintenance
Communication	Direct within the kernel	Through Inter-Process Communication (IPC)

f) State the different approaches of structuring an operating system.
1. **Simple Structure** (e.g., MS-DOS)
2. **Layered Approach** (Hierarchical layers)
3. **Microkernel** (Minimal kernel, services in user space)
4. **Hybrid Systems** (Combination of approaches, e.g., Windows)

Complete the following sentences:

a) GUI and CLI are types of user interface.
b) CPU and Memory are examples of operating system resources.
c) Command line interface needs Shell which accepts text inputs and converts commands to appropriate OS functions.

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d) Some systems have both GUI and CLI interface such as Windows and MAC OS.
e) Fork is a system call to create a new process.
f) System Programs are those programs that Provide a convenient environment for program development and execution.
g) The operating system is divided into a number of layers highest layer is user interface and the lowest is hardware.

multiple-choice questions (MCQs)

1. Which of the following is NOT a common type of user interface provided by an operating system?
   a) Command-Line Interface (CLI)
   b) Graphical User Interface (GUI)
   c) Batch Interface
   d) Kernel Interface
2. **Which method of passing parameters to a system call is the simplest and fastest but has limited capacity?**
   a) Passing parameters in a memory block
   b) Passing parameters on the stack
   c) Passing parameters in registers
   d) Passing parameters via message passing
3. **In a microkernel architecture, where are most operating system services, such as file systems and device drivers, typically run?**
   a) In the kernel mode as part of the monolithic kernel
   b) In the user mode as separate processes
   c) In the hardware layer
   d) In the system call interface
4. **What is a key characteristic of the MS-DOS operating system structure?**
   a) It uses a layered approach with strong modularity.
   b) It has a microkernel design.
   c) It is monolithic with no protection between user programs and the OS.
   d) All services run as separate user processes.
5. **What is the role of background services (daemons) in an operating system?**
   a) To provide a user interface
   b) To run automatically and perform essential background tasks
   c) To help compile and debug programs
   d) To edit file contents
6. **The "printf()" function in a C program eventually makes a call to which system call?**
   a) open()
   b) read()
   c) write()
   d) fork()

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State whether the following sentences are true or false:

a. Processes may exchange information either on the same computer or between computers over a network. **True**
b. If multiple users or multiple jobs running concurrently resources are necessarily allocated to each of them. **True**
c. System calls are always written in low level languages. **False**
d. System call are mostly accessed by application program interface rather than direct system calls. **True**
e. The caller of the system call must know how the system call is implemented. **False**
f. If the parameters to system call are passed through registers, this limits the number of parameters that can be passed to the system call. **True**
g. Kernel is one of the first components loaded into memory during the boot process. **True**
h. In virtual machine each process has its own processor and virtual memory. **False**

Best wishes
Dr. Samar A. Said

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